Highly spectrum-selective ultraviolet photodetector based on *p*-NiO/*n*-IGZO thin film heterojunction structure

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Abstract: Ultraviolet photodetector with *p*-*n* heterojunction is fabricated by magnetron sputtering deposition of *n*-type indium gallium zinc oxide (*n*-IGZO) and *p*-type nickel oxide (*p*-NiO) thin films on ITO glass. The performance of the photodetector is largely affected by the conductivity of the *p*-NiO thin film, which can be controlled by varying the oxygen partial pressure during the deposition of the *p*-NiO thin film. A highly spectrum-selective ultraviolet photodetector has been achieved with the *p*-NiO layer with a high conductivity. The results can be explained in terms of the "optically-filtering" function of the NiO layer.

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OCIS codes: (040.5160) Photodetectors; (040.7190) Ultraviolet; (310.0310) Thin films.

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1. Introduction

Photodetectors operating in the ultraviolet (UV) light range have many applications in the civil and military areas, such as flame detection, space-to-space communication, missile plume detection, astronomy and biological researches [1, 2]. Photodetectors based on narrow bandgap semiconductors, especially Si (bandgap E_g = 1.1 eV), have been commercialized for light detection for a long time. However, to realize the selective detection in the UV light range, an optical filter is usually needed to filter out the visible and infrared light, which greatly increases the complexity and cost of the photodetectors. To overcome this problem, wide bandgap semiconductors like ZnO, GaN, ZnMgO, $In_2Ge_2O_7$, Zn_2GeO_4, and ZnS have been investigated for UV light detecting application [3–9]. Compared with the Si-based photodetector, the photodetectors with wide bandgap semiconductors have many advantages like high breakdown field, filter-free UV light detection, high radiation-resistance, and highly chemical and thermal stabilities [10, 11]. There are three types of UV semiconductor photodetectors, including photoconductive detector, Schottky barrier photodetector, and *p-n*

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junction photodetector. Among them, photoconductive detector has attracted much attention due to its simple structure and high photo-responsivity from photoconductive gain; unfortunately, the high photoconductive gain is usually accompanied by the slow recovery speed, which can be attributed to the persistent photoconductive effect [12]. To avoid this problem, *p*-*n* junction photodetector is employed to realize fast response to the UV light. As compared to the photoconductive detector, the *p*-*n* junction photodetector has many advantages like low dark current, high impedance, capability for high-frequency operation, and good compatibility with planar-processing techniques; meanwhile, its low saturation current and high built-in voltage make it also superior to the Schottky barrier photodetector [13].

n-type indium gallium zinc oxide (*n*-IGZO) thin film has been investigated for UV light detection based on metal-semiconductor-metal or transistor structures [14–16]; and *p*-type nickel oxide (*p*-NiO) thin film has been used to form *p*-*n* junction UV detectors with *n*-type ZnO or *n*-type Si [17–20]. On the other hand, *p*-*n* heterojunction diode based on *p*-NiO and *n*-IGZO has been demonstrated [21, 22]. In this work, *p*-*n* heterojunction UV detector based on *n*-IGZO and *p*-NiO thin films has been fabricated. High spectrum-selectivity is achieved with the "optically-filtering" function of the *p*-NiO layer. The influence of the conductivity of the *p*-NiO thin film on the performance of the photodetector has been also investigated in this work.

2. Experiment

The *p*-NiO/*n*-IGZO heterojunction photodetector was fabricated with the following sequence. Firstly, IGZO thin film with the thickness of 170 nm was deposited on a commercial ITO glass by radio frequency (RF) magnetron sputtering of an IGZO target in Ar ambient. Circular patterns with the diameter of 300 μm were defined by lithographic process. Then, a 130 nm NiO thin film was deposited by RF sputtering of a NiO target in a mixed Ar/O_2 ambient. Low conductivity (L-NiO), medium conductivity (M-NiO), and high conductivity (H-NiO) of the NiO thin films were achieved by sputtering at the oxygen partial pressure of 0, 6×10^{-5}, and 1×10^{-4} Torr, respectively. After the growth of NiO layer, ITO thin film as the top electrode layer was deposited by RF sputtering. Finally, lift-off process was carried out. Carrier concentrations of the IGZO and NiO thin films were measured with a Hall effect measurement system (Nanometrics HL5500PC). Optical transmittance of the thin films was measured with an UV-Vis spectrophotometer (Perkin-Elmer 950) in the wavelength range of 250 - 900 nm. Electrical characteristics and photoresponse spectra of the as-fabricated photodetectors were measured with a Keithley 4200 semiconductor characterization system, a 300 W Xenon light source and an UV-Vis-IR monochromator.

3. Results and discussion

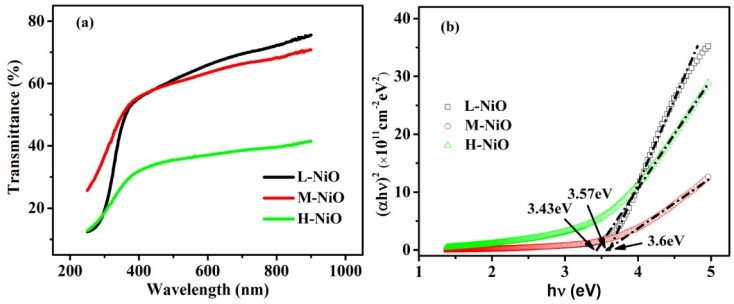


Fig. 1. (a) Optical transmittance spectra of the L-NiO, M-NiO, and H-NiO thin films; (b) Plots of $(\alpha hv)^2$ versus hv for the L-NiO, M-NiO and H-NiO thin films.

Electrical properties of the IGZO and NiO thin films were measured with the Hall effect measurement system in the Van der Pauw configuration at room temperature. The carrier

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concentrations of the IGZO, H-NiO, and M-NiO thin films obtained from the Hall effect measurement are 4.4 × 10^{19} cm^{-3} (electrons), 2.8 × 10^{19} cm^{-3} (holes), and 3.1 × 10^{17} cm^{-3} (holes), respectively. Due to the high resistivity, reliable data cannot be obtained for the L-NiO thin film. To examine the optical properties of the NiO thin films, 80 nm NiO thin films with different conductivities were deposited on quartz glass substrates, respectively. The optical transmittance is measured in the wavelength range of 250 - 900 nm. As observed in Fig. 1(a), the average transmittance of the L-NiO, M-NiO, and H-NiO thin films in the visible range (400 - 700 nm) are 63.25%, 61.62%, and 35.98%, respectively, indicating a decreasing trend of the transmittance with the conductivity of the NiO thin film, which can be attributed to the increasing concentration of the intrinsic defects in the films. With the increase of the oxygen partial pressure during the sputtering deposition, more nickel vacancies, acting as acceptors in the *p*-NiO films, are created accompanying the formation of Ni^{3+} ions, which exhibit charge transfer transitions in the oxide matrix resulting in a strong, broad absorption band in the visible range [23, 24]. Meanwhile, stress field induced by the intrinsic defects may cause the light scattering effect [25]. In addition, free-carrier absorption may also occur in the long-wavelength region (e.g., the near infrared region) [26]. Thus, the transmittance will decrease with the conductivity of the NiO thin film. The optical bandgap of the NiO thin film can be estimated with

$$\alpha h\upsilon = A(h\upsilon - E_g)^m \tag{1}$$

where α is the optical absorption coefficient, A is a constant, E_g is the optical bandgap, $h\upsilon$ is the photon energy, and m is 1/2 for direct bandgap [27]. The absorption coefficient α is calculated with

$$\alpha \approx \ln(1/T)/d \tag{2}$$

where T is the transmittance and d is the thin film thickness [28]. As shown in Fig. 1(b), the direct bandgaps for the L-NiO, M-NiO, and H-NiO films are 3.60 eV, 3.57 eV, and 3.43 eV, respectively, which are within the reported bandgap range for *p*-NiO film (3.4 - 3.8 eV) [29]. The bandgap of IGZO thin film determined with spectroscopic ellipsometry is ~3.45 eV as reported in our previous work [30]. The wide bandgaps of both IGZO and NiO films promise a low response of the photodetector to visible and infrared light.

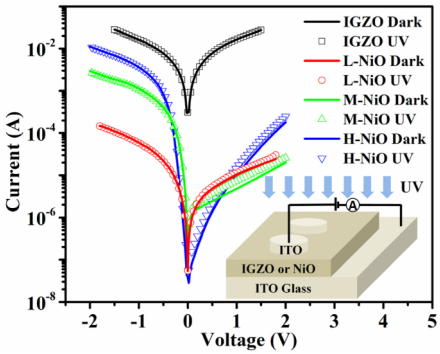


Fig. 2. I-V characteristics of the ITO/IGZO/ITO, ITO/L-NiO/ITO, ITO/M-NiO/ITO, and ITO/H-NiO/ITO thin film structures in the dark or under 365 nm UV light illumination. The inset shows the schematic illustration of the thin film structures.

To examine the photoelectric response to UV light of the IGZO (or L-NiO, M-NiO, H-NiO) thin film itself, a simple ITO/IGZO (or L-NiO, M-NiO, H-NiO)/ITO thin film structure was also fabricated, as schematically illustrated in the inset of Fig. 2. Symmetric current-voltage (I-V) curve is observed for the IGZO-based structure due to the high conductivity of the IGZO thin film itself. The asymmetric I-V curves of the NiO-based structures mainly

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originate from the difference in the quality between the top sputtered ITO electrode and bottom commercial ITO electrode. As can be seen in Fig. 2, the UV illumination doesn't produce a significant influence on the I-V curves of all the four structures. This indicates that the UV-induced relative change in the carrier concentration of the IGZO (or L-NiO, M-NiO, H-NiO) thin film is insignificant and the thin film structures do not have the capability of UV light detection.

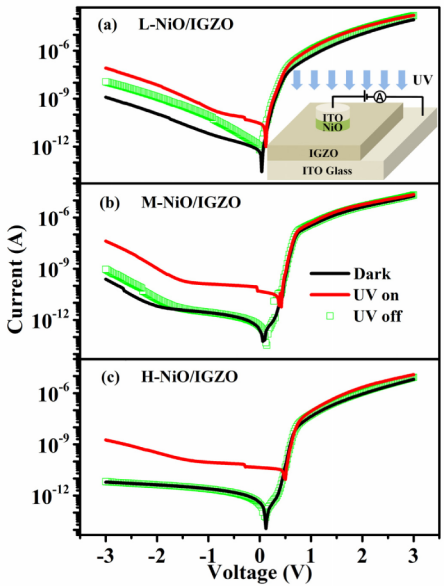


Fig. 3. I-V characteristics of (a) ITO/L-NiO/IGZO/ITO, (b) ITO/M-NiO/IGZO/ITO, and (c) ITO/H-NiO/IGZO/ITO structures measured under the following sequent conditions: dark, 365 nm UV light illumination, and UV light off. The inset in (a) shows the schematic illustration of the ITO/*p*-NiO/*n*-IGZO/ITO structures.

In contrast to the above simple thin film structures, with the formation of the *p*-NiO/*n*-IGZO heterojunction, the ITO/*p*-NiO (L-NiO, M-NiO, or H-NiO)/*n*-IGZO/ITO structures show an obvious electrical rectification behavior and a large photoelectric response to the UV illumination, as shown in Fig. 3(a)-3(c). The reverse dark current measured at −3 V for the structures with L-NiO, M-NiO, and H-NiO thin films are 1.23×10^{-9} A, 2.47×10^{-10} A, and 6.15×10^{-12} A, respectively. This shows the typical behavior of a *p*-*n* junction, i.e., a higher acceptor concentration in *p*-NiO layer leads to a lower reverse current. Obviously, to have a high signal to noise ratio, a low dark current is required; thus the H-NiO layer is more suitable for UV light detection. For all the three structures with different *p*-NiO layers (i.e., L-NiO, M-NiO, or H-NiO), the UV illumination causes an increase in the reverse current by about two orders, showing a promising application in UV light detection.

As shown in Fig. 3, the reverse currents of the structures with L-NiO or M-NiO cannot fully recover back to their original levels after the UV light is off; in contrast, the reverse current of the structure with H-NiO is fully recovered after the UV light is off. The phenomena could be attributed to the effect of the UV-induced hole trapping in the *p*-NiO layer. UV illumination produces electron-hole pairs in both *p*-NiO and *n*-IGZO layers. If some of the UV-generated holes are trapped in the deep-levels in the *p*-NiO layer, the I-V characteristic of the *p*-*n* junction would be affected by the hole trapping. The hole trapping would partially compensate the negative space charge in the *p*-NiO side of the depletion region of the *p*-*n* junction, reducing the built-in electric field as well as the barrier height of the *p*-*n* junction. This would have a more significant impact on the small reverse current than on the large forward current. Compared to H-NiO, L-NiO and M-NiO have a lower

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concentration of acceptors, thus their densities of the negative space charge are lower and the widths of the depletion region in the *p*-NiO are larger. Therefore, the charge compensation and its effect on the reverse current are more significant for L-NiO and M-NiO than for H-NiO. This explains the difference in the recovery of the I-V characteristic (in particular the reverse current) after UV light is off among the photodetector structures with different *p*-NiO conductivities. Obviously, the full recovery of the structure based on H-NiO as shown in Fig. 3(c) makes the structure suitable as an UV photodetector.

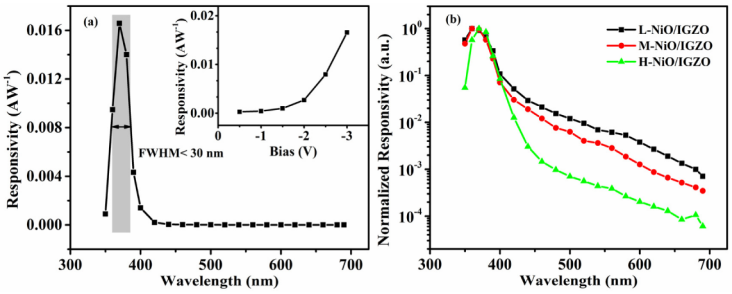


Fig. 4. (a) Spectral responsivity of the ITO/H-NiO/IGZO/ITO photodetector under the bias of −3 V. The inset shows the responsivity as a function of the reverse bias; (b) Normalized spectral responsivities of the ITO/L-NiO/IGZO/ITO, ITO/M-NiO/IGZO/ITO, and ITO/H-NiO/IGZO/ITO photodetector structures under the bias of −3 V.

Figure 4(a) shows the spectral responsivity of the photodetector structure based on H-NiO at −3 V bias. A peak responsivity of 0.016 A/W is observed at the wavelength of ~370 nm with the full width at half maximum (FWHM) of smaller than 30 nm, showing a high spectrum selectivity. The peak responsivity of 0.016 A/W is comparable or better than that of some of the metal-oxide-based *p*-*n* junction UV detectors reported in literature [3, 17, 31–33] Note that the responsivity of the photodetector in this work can be further improved by optimizing the thicknesses of the ITO top electrode, *p*-NiO and *n*-IGZO layers. Figure 4(b) shows the normalized responsivity of the photodetectors with L-NiO, M-NiO, and H-NiO thin films. It can be concluded from the figure that the photodetector based on the H-NiO thin film has the best spectral selectivity, which is explained in the following. The transmittance of the ITO top electrode decreases sharply when the wavelength is shorter than ~400 nm, which should be partially responsible for the falling of the responsivity at the wavelengths shorter than ~370 nm. However, the ITO top electrode with the same thickness is used in all the three structures. This suggests that the difference in the spectral responsivity among the structures is related to the difference in the transmittance of the NiO thin film. For the light with the wavelengths shorter than ~360 nm, the photon energy is larger than the bandgap of the NiO film; thus most of the photons arriving in the NiO film are absorbed in the neutral region of the top NiO layer instead of reaching the depletion region [33]. In this way, the top NiO layer works as a "filter" to filter out the light with short wavelengths. For the visible and infrared light, though the photons can reach the depletion region, they cannot excite electron-hole pairs due to their energies smaller than the bandgap of NiO and IGZO films; thus low responsivity is obtained. As H-NiO film has the lowest near UV-visible-near infrared transmittance, as shown in Fig. 1(a), the photodetector based on H-NiO thus has the best spectral selectivity. As can be observed in Fig. 4(b), among the three photodetector structures, the H-NiO/IGZO photodetector has the highest UV to visible rejection ratio (e.g., the ratio of responsivity at the wavelength of 370 nm to the responsivity at 500 nm is 1030 for the H-NiO/IGZO photodetector, while it is only 60 for the L-NiO/IGZO photodetector) due to the lowest visible transmittance of H-NiO film, showing the best visible blindness. This is important for the application of an UV photodetector in a visible light background [34]. It can be also observed from Fig. 4(b) that the wavelength of the peak responsivity shifts from ~370 nm (H-NiO) to ~360 nm (L-NiO) (blue shift) with the decrease of the conductivity of NiO film. The blue shift is due to the bandgap increase of the *p*-NiO films (note that the direct bandgaps for L-

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NiO, M-NiO, and H-NiO films are 3.60 eV, 3.57 eV, and 3.43 eV, respectively), as shown in Fig. 1(b). On the other hand, the influence of the reverse bias on the responsivity of the H-NiO/IGZO photodetector has been examined, and the result is shown in the inset of Fig. 4(a). With the increase of the reverse bias, a larger photocurrent is produced due to the widening of the depletion region, and thus the responsivity increases.

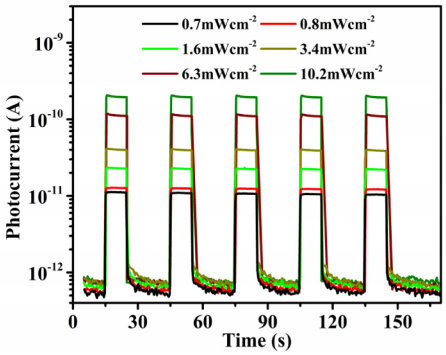


Fig. 5. Experiment on the repeatability and photocurrent response of the H-NiO/IGZO photodetector under the bias of –0.2 V, at the wavelength of 365 nm and with various UV light intensities.

Good repeatability and fast response are critical to the detection of a quickly varying UV signal. An experiment on the repeatability and photocurrent response was carried out on the H-NiO/IGZO photodetector at the wavelength of 365 nm with various UV light intensities. The UV light was mechanically switched on for 10 s and off for 20 s alternatively. The result is shown in Fig. 5. As can be observed in the figure, good repeatability and fast response have been achieved, in a wide light intensity range of 0.7 - 10.2 $mWcm^{-2}$.

4. Conclusion

In conclusion, *p*-NiO/*n*-IGZO heterojunction can be used to realize an UV photodetector. The performance of the photodetector is largely affected by the concentration of acceptors in the *p*-NiO layer, which can be controlled by varying the oxygen partial pressure during the sputtering deposition of the *p*-NiO layer. A highly spectrum-selective UV photodetector has been achieved with the *p*-NiO layer with a high concentration of acceptors. The photodetector has a good repeatability and fast response.

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